Trust-Calibrated Multi-Agent Scientific Deliberation for Mitigating Sycophantic Consensus in LLM Reasoning

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Abstract

Large language models now answer hard scientific questions, yet they still hallucinate and follow broken lines of reasoning. Multi-agent debate offers one way to push back on this. When several models argue a question out before committing to an answer, the reasoning usually gets sharper and errors get caught.

The weakness is in how these debates settle. Most systems pick a winner by majority vote or by which agent sounds most confident. When a confident majority is simply wrong, it can pull a correct minority agent off its answer, and the group lands on the wrong conclusion. No current method checks whether an agent's claims actually hold up against outside scientific evidence before letting that agent sway the room.

We address this with a Trust-Calibrated Multi-Agent Scientific Deliberation framework that adjusts how much each agent counts as the debate unfolds.

The framework breaks every agent response into small factual claims. For each claim it retrieves relevant scientific literature and checks whether the evidence backs the claim or contradicts it. Every agent then carries a trust score that moves up or down as these checks accumulate over the rounds of debate. The final answer is built from trust-weighted aggregation rather than a raw count of votes.

Tying trust to real evidence means unsupported claims lose weight, so a correct but outnumbered agent is far less likely to be silenced.

We expect the approach to cut hallucination and keep reasoning tied to verifiable sources. It should also stop the debate from settling too early on a wrong answer, giving more reliable scientific results, and it does this without any model fine-tuning.

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Publication List

The main contributions of this research are either published or accepted or in preparation in journals and conferences as mentioned in the following list:

Journal Articles

1. Leveraging Machine Learning in Balancing Inventories and Analyzing Market Trends: A Comprehensive Systematic Review (Artificial Intelligence Review, Under processing)

Conference Papers

1.

Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

1.

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Chapter 1

Introduction

This chapter introduces the project. It presents the background and problem, the motivation behind the work, the objectives, a short summary of the methodology, the expected outcome, and the structure of the rest of the report.

1.1 Project Overview

Large language models now handle many reasoning tasks, from scientific question answering to tutoring and decision support. A single model can still be confidently wrong. It may state an incorrect answer in fluent and assured language, which makes the mistake hard to catch. Multi-agent debate was proposed to reduce this problem. Several agents answer the same question, read each other's reasoning, and revise their positions over a few rounds before a final answer is selected [1]. Chain-of-thought prompting gives each agent a way to expose its intermediate steps [2], which makes the reasoning inside a debate easier to inspect. The real difficulty is the final step, where the separate answers are combined. Most debate systems settle disagreements by majority vote. Every agent gets equal weight, and the most common answer wins. This is simple, but it treats popularity as a stand-in for correctness. In scientific reasoning the better answer is the one backed by evidence, not the one repeated by more agents. The study examines a trust-calibrated multi-agent deliberation framework that adjusts each agent's influence during the debate based on how well its claims match retrieved external evidence. The final answer then depends on evidence-grounded trust rather than a raw headcount.

1.2 Motivation

Majority voting can fail in a way that is easy to miss. A debate may look like it reached agreement while it has quietly dropped a correct answer held by one agent. Recent studies of multi-agent debate report that agents often copy or switch answers under social pressure instead of reasoning on their own, and that the correct answer is present in the discussion but ignored in more than twenty percent of wrong-answer cases [3]. Work on model behaviour points to a related tendency, where models trained to be agreeable grow more sycophantic and less accurate [4]. When a confident majority pushes a correct minority agent to give up its answer, majority voting does more than pick the wrong result. It hides the failure behind an appearance of consensus. This matters most for scientific questions, where a wrong but confident consensus can mislead a user into trusting an answer that deserved a closer check. Existing methods approach this failure from different directions. iMAD decides when a debate is worth running [5], DebUnc uses self-reported confidence to adjust agent influence [6], and Mixture-of-Agents combines outputs through static aggregation [7]. These methods are useful, yet none ties an agent's influence to whether its claims hold up against external evidence while the debate is still active. That gap is what motivates the study.

1.3 Objectives

The main objective of this project is to reduce sycophantic consensus in multi-agent LLM debate by replacing plain majority voting with evidence-grounded trust calibration. This objective involves studying how majority-vote debate fails when a confident wrong majority pressures a correct minority agent into changing its answer, designing a bounded trust score that rises when an agent's claims are supported by external evidence and falls when they are contradicted, and applying that score during the debate before the final answer is fixed. The work also compares the proposed method with standard majority voting and other relevant debate and aggregation baselines, then evaluates the framework using answer accuracy together with consensus-collapse, minority-preservation, and evidence-calibration metrics.

1.4 Methodology

The study follows an experimental methodology. A baseline multi-agent debate system is built first, where several heterogeneous LLM agents answer the same scientific question, share their reasoning, and revise across rounds. The baseline uses majority voting so the usual failure mode can be observed and measured. Each agent's reasoning is then broken into smaller factual claims. These claims are checked against scientific evidence retrieved from external sources such as academic papers and reference databases. Any contradiction between a claim and the retrieved evidence is detected and scored [8], building on the idea that external tools can verify and correct model output [9]. The verdicts are turned into trust updates. Supported claims raise an agent's trust, contradicted claims lower it, and uncertain claims are handled conservatively. The trust score is bounded so that no agent gains unlimited influence or drops out of the debate completely. The final answer is chosen through trust-weighted aggregation rather than a simple count. The framework is evaluated on scientific reasoning datasets such as GPQA and MMLU-Pro. The evaluation checks whether the mechanism lowers consensus collapse, keeps correct minority answers, and holds overall accuracy without any model fine-tuning.

1.5 Project Outcome

The main outcome is a reproducible framework for trust-calibrated multi-agent scientific deliberation. It shows how retrieved evidence can update agent influence during a debate, and how the final answer can be decided by evidence-weighted trust instead of a majority vote. Alongside it, the study produces an evaluation protocol that detects sycophantic consensus, measures how often correct minority answers survive, and compares the method against established baselines. The study is expected to show whether dynamic trust calibration makes multi-agent debate more dependable on scientific questions. Even where the method does not remove every wrong answer, it should make the decision more transparent by recording which agents were supported or contradicted by the evidence.

1.6 Organization of the Report

The rest of this report is arranged as follows. Chapter 2 covers the background, the related work on multi-agent debate and sycophancy, and the gap this project addresses. Chapter 3 presents the requirement analysis, the detailed methodology, the system design, and the project plan. Chapter 4 describes the implementation setup, the testing and evaluation procedure, and the results with discussion. Chapter 5 discusses the relevant standards, the impact of the work, sustainability, and the complex engineering problem analysis. Chapter 6 concludes the report, notes its limitations, and gives recommendations for future work.

Chapter 2

Background

This chapter introduces the concepts needed to understand the proposed system and reviews recent work on multi-agent debate, sycophancy, and agent influence. The last part compares the reviewed methods and explains the gap addressed by this project.

2.1 Preliminaries

Large language models (LLMs) generate answers one token at a time, but their outputs can still contain unsupported or incorrect reasoning. Fluent wording does not guarantee that an answer is correct. Chain-of-thought prompting asks a model to expose intermediate reasoning steps before giving a final answer, which makes the reasoning easier to inspect and gives later components more information to evaluate [2]. These steps are useful evidence about how an answer was formed, but they should not be treated as proof of correctness by themselves. Multi-agent debate extends this process by asking several agents to solve the same problem independently. The agents then receive the other answers and explanations, discuss their differences, and revise their positions over a fixed number of rounds. A final answer is selected after the discussion [1]. Several agents are useful because their independent attempts can expose different errors and allow one agent to question another agent's reasoning. The way those answers are combined determines whether debate helps. A common method is majority voting, where each agent has the same weight and the most frequent answer is selected. This rule is easy to implement, but it assumes that the majority is more likely to be correct. If several agents repeat the same mistake, the vote can strengthen the mistake instead of correcting it. A debate can therefore reach agreement without reaching a correct answer. This failure becomes more serious when agents change their answers because of peer pressure rather than new evidence. In this report, *sycophantic consensus* refers to the tendency of an agent to copy or exchange answers with other agents and to abandon a correct position when a wrong position has greater social support. This behaviour is different from a justified revision. A change based on a retrieved source or a valid counterargument can improve the result, whereas a change based only on the confidence or number of supporting agents can remove the correct minority answer. Trust and confidence are related but not identical. Confidence may be provided by an agent in its response, or estimated from properties of its token distribution. Both forms describe a signal produced by the model. They do not directly verify whether the claim is true. DebUnc makes this distinction clear by comparing deployable uncertainty measures with a diagnostic Ground Truth measure that uses the correct answer. The diagnostic measure performs much better, but it cannot be used during normal deployment because the answer is not available to the system in advance [6]. The framework uses this distinction to define trust more narrowly. An agent should gain influence when its claims are supported by retrieved scientific evidence and lose influence when its claims are contradicted. A lack of evidence is not treated as a contradiction. This allows the system to separate an agent that is unsupported from one whose claims conflict with the available literature. Zhang et al. (2025) provide a survey of reasoning-oriented LLMs, which gives broader context for the reasoning methods discussed in this report [10].

2.2 Literature Review

The review focuses on three recent papers that cover different parts of the problem. iMAD studies when a debate should be triggered, CONSENSAGENT studies sycophancy in multi-agent debate and changes the task prompt when it is detected, and DebUnc changes the influence of agents during a debate using uncertainty signals. Together, these papers provide the main comparison points for the proposed framework.

2.2.1 Similar Applications

The system is not a conventional web or mobile application. It is an experimental deliberation layer that could support a scientific question-answering tool, a research assistant, or another decision-support interface. Its closest similar systems are therefore multi-agent LLM applications that generate several candidate answers and combine them before returning a result. The important difference is that this project exposes the evidence checks and the changing trust of each agent as part of the result, rather than treating the final answer as a simple vote.

2.2.2 Related Research

Wei et al. (2022) showed that chain-of-thought prompting can improve the performance of large language models on tasks that require multi-step reasoning [2]. Their method asks the model to produce intermediate reasoning steps before the answer. These traces make the reasoning easier to inspect, but they do not guarantee that the reasoning is sound. Du et al. (2024) extended this idea through multi-agent debate, where several agents answer a question, read each other's reasoning, and revise their positions over multiple rounds [1]. Their work showed how communication between agents can improve factuality and reasoning, although the final decision still depends on how the separate answers are combined. Gou et al. (2024) proposed CRITIC, a tool-interactive framework that uses external tools such as search engines, code interpreters, and toxicity classifiers to check and improve model responses [9]. The framework showed that external verification can help models correct some errors, but it did not involve several models debating the same question. Wang et al. (2025) introduced Mixture-of-Agents, which combines outputs from several proposer models through layered aggregation and synthesis [7]. Their results demonstrated the value of combining different LLM outputs without fine-tuning, but the aggregation remained static and did not maintain an explicit trust value for each agent or use external scientific evidence. Fan, Yoon, and Ji (2026) proposed iMAD to reduce the cost of running debate on every question [5]. Their method asks one agent to produce an initial reasoning path, an alternative answer, and confidence scores, then extracts 41 linguistic and semantic features from the response. A multilayer perceptron predicts whether a debate is likely to correct the first answer. When the score falls below a selected threshold, a three-agent debate is started. Across six question-answering and visual question-answering datasets, the paper reports token savings of up to 92 percent compared with GroupDebate and an accuracy improvement of up to 13.5 percentage points over the single-agent chain-of-thought baseline. However, iMAD does not change how agents are trusted after the debate begins, because standard aggregation is still used. Pitre, Ramakrishnan, and Wang (2025) examined sycophancy as a failure mode inside multi-agent debate [3]. They reported that agents may copy or swap answers instead of continuing independent reasoning, and that the correct answer is present in the discussion but ignored in more than 20 percent of wrong-answer cases. Their CONSENSAGENT framework detects stalling, answer swapping, and high similarity between explanations. A fine-tuned GPT-4o model then rewrites the task prompt before the agents debate again, after which confidence and consistency are used in the final aggregation. The paper reported a reduction in measured sycophancy between 7 and 30 percent across its evaluation settings. This method is close to the present problem, but it mainly treats prompt ambiguity and still relies on self-reported confidence rather than claim-level evidence. Yoffe, Amayuelas, and Wang (2025) proposed DebUnc to share uncertainty information between agents during debate [6]. They measured each agent using Mean Token Entropy or TokenSAR and communicated the resulting confidence through the prompt or through attention scaling. On the Llama-3 evaluation reported in the paper, average accuracy moved from 0.63 for standard debate to 0.64 with the best deployable attention setting, while a diagnostic Ground Truth signal raised the average to 0.73. This result suggests that the quality of the trust signal is a limiting factor. The diagnostic signal requires the correct answer and cannot be used during normal deployment, while attention scaling requires access to model internals. Kushwaha and Madhavan (2026) described an eight-agent MAS-RAG architecture for finding contradictions among retrieved documents and ranking evidence by credibility [8]. Their method is relevant because it brings external evidence into the reasoning process, but the credibility of each source is fixed at the beginning by a human curator and is not updated during deliberation. Cau et al. (2025) presented LODAS and examined how opinions converge in LLM interactions [11]. They found that convergence is often caused by selective persuasion rather than simple agreement and also showed that weak arguments can mislead other agents. The study did not provide a method for checking whether facts used in those arguments were correct. Yeow et al. (2025) studied reliable decision-making in multi-agent LLM systems [12]. They compared majority voting, averaging, decentralized configurations, and iterative feedback. Their findings indicate that independent decision architectures can be more reliable, whereas feedback between agents may allow early errors to spread through later rounds. Wang et al. (2024) examined whether multi-agent discussions extend the reasoning ability of a single LLM [13]. They found that a single model with well-designed in-context examples can perform comparably to multi-agent discussion systems, while the main benefits of debate appear in zero-shot settings. They also identified judge errors and peer conformity as weaknesses of multi-agent deliberation. Choi, Zhu, and Li (2025) separated communication from voting to study where the gains from debate originate [14]. Their analysis suggests that majority voting accounts for much of the reported improvement, while unguided discussion does not necessarily move agent beliefs toward the correct answer. This finding supports the need for an explicit intervention that gives more influence to evidence-backed claims. Cui et al. (2026) introduced FREE-MAD, a consensus-free debate framework that uses anti-conformity prompts and trajectory-based scoring instead of final-round consensus [15]. Evaluated on eight reasoning benchmarks, the method reportedly reduced token use by about 50 percent and limited error propagation. It addresses conformity by avoiding consensus, but it does not update agent influence through external scientific verification. Zhou et al. (2025) proposed ReSo, a reward-driven self-organizing multi-agent system that builds dynamic task execution graphs and uses a Collaborative Reward Model to select agents at different stages [16]. Their reported results on Math-MAS and SciBench-MAS suggest that adaptive agent selection can help on tasks where fixed multi-agent structures perform poorly. Ibrahim et al. (2026) examined the effect of tuning language models to sound warm and agreeable [4]. Across several model families, they reported lower factual accuracy and greater susceptibility to sycophancy after warmth-oriented tuning, including error increases of 10 to 30 percentage points in some domains. This result gives background to the social-pressure problem in multi-agent debate. Taken together, these studies show that current systems can improve reasoning through multiple agents, external tools, uncertainty signals, prompt changes, and adaptive coordination. They also leave a specific gap. The reviewed methods do not continuously update an agent's influence during scientific deliberation according to whether its claims are supported by retrieved evidence. The proposed framework addresses this gap by combining multi-round debate with claim-level evidence checking and a bounded trust score.

2.3 Gap Analysis

The reviewed papers leave three needs in multi-agent debate. A system must decide whether the extra cost of debate is justified, detect when a discussion is moving toward unsupported agreement, and decide which agent should have more influence when the agents disagree. iMAD mainly addresses the first need. CONSENSAGENT addresses the second through prompt optimization. DebUnc addresses the third with an in-debate weighting mechanism, but its practical signals are internal uncertainty estimates. The remaining gap is an in-session trust update based on independently checkable evidence. The framework addresses this gap by decomposing an agent's response into smaller claims, retrieving relevant scientific passages, and classifying each claim as supported, contradicted, or unverifiable. The verdicts update a bounded trust score after each debate round. An agent can therefore be in the numerical majority and still lose influence if the claims supporting its position are contradicted. A correct minority agent can retain influence when its claims are better supported. The signal is not supplied by the agent that benefits from the influence, and the update happens before the final answer is fixed, so it can affect later revisions as well as the final aggregation. The design does not assume that retrieval is perfect. No evidence is recorded as an unresolved case, while direct contradiction is handled separately. This distinction is needed because absence of a retrieved source should not be treated as proof that a claim is false. The gap also affects evaluation. The reviewed papers report accuracy and, in different settings, token use, debate rounds, consensus, or copying behaviour. Those measures are useful but do not directly show whether a correct minority agent was pressured into changing its answer. The evaluation therefore uses consensus-collapse rate (CCR), minority-preservation rate (MPR), and evidence- calibration rate (ECR) alongside ordinary answer accuracy. CCR measures whether correct agents abandon their initial answer under injected pressure without new evidence. MPR measures whether a correct minority position survives to the final output. ECR measures whether the trust scores correspond to the agents' verified correctness. The contribution is narrower than a general solution to LLM hallucination. It targets the decision process inside scientific multi-agent debate. Its main claim is that agent influence should depend on externally checked support, not only on the number of agents taking a position or the confidence with which that position is stated. Retrieval sparsity, imperfect claim decomposition, and correlated errors between agents remain important limitations and will be examined during the evaluation.

2.4 Summary

This chapter introduced LLM reasoning, chain-of-thought prompting, multi-agent debate, and sycophantic consensus. The literature review showed that iMAD reduces the cost of deciding when to debate, CONSENSAGENT mitigates problematic debate through prompt optimization, and DebUnc changes in-debate influence using internal uncertainty.

Their results support the need for a better trust signal, but none of these methods updates agent influence from retrieved external scientific evidence. The next chapter uses this gap to define the system requirements, detailed methodology, and project plan.

Chapter 3

Project Design

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

3.1 Requirement Analysis

3.1.1 Functional and Nonfunctional Requirements

3.1.2 Context Diagram

3.1.3 Data Flow Diagram Level 1

3.1.4 UI Design

3.2 Detailed Methodology and Design

You have to mention alternate solutions that you have considered. Why you have selected the specific solution, etc.

3.3 Project Plan

3.4 Task Allocation

3.5 Summary

Chapter 4

Implementation and Results

[Must be present in Final Report. Incomplete version might be included in FYDP-1 Report, however it is optional.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

4.1 Environment Setup

4.2 Testing and Evaluation

4.3 Results and Discussion

4.4 Summary

Chapter 5

Standards and Design Constraints

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

5.1 Compliance with the Standards

Only mention the standards that are related to your project. This list is not complete. For each of the standards discuss the alternates with pros and cons and rationale of selection.

5.1.1 Software Standards

5.1.2 Hardware Standards

5.1.3 Communication Standards

5.2 Design Constraints

Only mention the constraints that are related to the design of your project. This list is not complete.

5.2.1 Economic Constraint

5.2.2 Environmental Constraint

5.2.3 Ethical Constraint

5.2.4 Health and Safety Constraint

5.2.5 Social Constraint

5.2.6 Political Constraint

5.2.7 Sustainability

5.3 Cost Analysis

Provide a cost analysis in terms of budget required and revenue model. In case of budget, you must show an alternate budget and rationales.

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 5.1). For P1, you need to put another mapping with Knowledge profile and rational thereof.

Table 5.1: Mapping with complex problem solving.

P1 Dept of Knowledge	P2 Range of Conflicting Requirements	P3 Depth of Analysis	P4 Familiarity of Issues	P5 Extent of Applicable Codes	P6 Extent of Stakeholder Involvement	P7 Interdependence
√	√					

5.4.2 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 5.2).

5.5 Summary

Table 5.2: Mapping with complex engineering activities.

A1 Range of resources	A2 Level of Interaction	A3 Innovation	A4 Consequences for society and environment	A5 Familiarity
$\surd$	$\surd$			

Chapter 6

Conclusion

[Must be present in FYDP-1 Report and also in Final Report. Might be incomplete in FYDP-1 Report.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

6.1 Summary

6.2 Limitation

6.3 Future Work

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